# Analyzing Physical Infrastructure, Operational Capacity, and Community Reach of U.S. Public Libraries: Evidence from the FY 2018 Public Libraries Survey

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***Abstract***—Public libraries are central to how Americans find and use information. Yet we know surprisingly little about how their buildings and hours of service actually differ across the country. This paper analyzes the FY 2018 Public Libraries Survey (PLS) outlet-level dataset from the Institute of Museum and Library Services (IMLS)—17,478 outlets across all 50 states, D.C., and U.S. territories—to document these patterns. Drawing on descriptive statistics, cross-tabulation, correlation analysis, and regression visualization, we examine outlet type distributions, geographic concentration, facility sizes, operating schedules, and the relationship between county population and service capacity. Central and branch libraries account for roughly 94% of all outlets. City libraries maintain longer hours and occupy larger buildings than rural ones. Suburban areas host the most outlets nationwide. County population shows a consistent positive relationship with both annual operating hours and building size. Over 82% of outlets stay open year-round. Together, these findings reveal a measurable service capacity gap between urban and rural areas—one with direct implications for how library resources should be allocated.

Dataset downloaded and used from https://catalog.data.gov/dataset/public-library-survey-pls-2018 (data.gov)

***Keywords***—public library infrastructure; IMLS public libraries survey; urban-rural library disparity; library operating hours; outlet type distribution; library service equity; NCES locale classification; county population.

## I. INTRODUCTION

Public libraries sit at an unusual intersection in American civic life. They are free, open to all, and ask nothing of visitors beyond returning what they borrow. As Klinenberg argued, they function as social infrastructure—physical spaces whose size, location, and hours shape community cohesion, democratic participation, and access to information [3]. That physical dimension matters: a library too small to hold its community or open too few hours to catch working adults cannot do its job, no matter how good its collection or staff [4].

Despite this, a national-scale analysis of how building size and operating hours vary by outlet type, urbanization level, and local population has remained thin in the literature. Studies have looked at digital inclusion [5], [6] and income-based service gaps using earlier PLS data [7], [8]. But the specific relationships among square footage, annual and weekly hours, outlet category, locale type, and county population—examined together and at the full national scale—have not been worked out. This study fills that gap using the FY 2018 PLS outlet-level public use file, which covers 17,478 individual library outlets. The FY 2018 data is the last complete pre-pandemic snapshot before COVID-19 disrupted operations, giving us a clean baseline of the system at normal running capacity [9].

Four research questions organize the analysis:

*RQ1: How are library outlets distributed across outlet types and geographic locales?*
*RQ2: What do national distributions of physical size and annual operating hours look like?*
*RQ3: To what degree do physical size and operating hours differ across locale categories?*
*RQ4: What is the relationship between county population and library operational capacity?*

The paper proceeds as follows. Section II situates the study within prior work on library service equity and infrastructure. Section III describes the data, preprocessing, and methods. Section IV presents results. Section V discusses findings and policy implications. Section VI concludes.

## II. LITERATURE REVIEW

### *A. Public Libraries as Social Infrastructure*

Treating the public library as infrastructure rather than just a service has a relatively recent scholarly history. Klinenberg's Palaces for the People reframed libraries as physical anchors of civic life: their spatial distribution and operating capacity affect community resilience and social trust [3]. The American Library Association has made a similar point—equity of access requires not only removing financial barriers but actively matching facility size and schedules to the heterogeneous needs of different communities [1]. This shift in framing moves policy discourse away from collections alone and toward physical presence: how big is the building, when is it open, and can people actually get there?

### *B. Service Disparities Along Income and Urbanization Lines*

The empirical record on library service disparities is clearest along income and urbanization lines. Sin's national study, using PLS data alongside U.S. Census tract information, found that libraries in lower-income or rural neighbourhoods consistently offer lower per-capita service across nearly every dimension—smaller buildings, shorter hours, fewer staff, smaller collections [7]. The estimated effect was not trivial: a one-percent increase in urbanization level was associated with a 0.71% larger outlet [8]. Real, Bertot, and Jaeger extended this to digital inclusion specifically, finding that rural libraries operate at a smaller scale with lower broadband capacity and less ability to support government service delivery—problems driven by structural funding limitations, not staff effort [10]. Local property-tax financing compounds these gaps by directing more dollars to already-affluent communities [7], [11].

### *C. The IMLS Public Libraries Survey as a Research Instrument*

The IMLS PLS is the only annually updated, census-level dataset on U.S. public library outlets [12]. Running since 1988, with response rates above 97% since FY 2010, it captures outlet-level data on building dimensions, operating schedules, collections, technology access, staffing, and finances [9], [13]. It has been used for service equity studies [7], rural library classification [14], and population health analyses [15]. The NCES locale classification embedded in the PLS—grouping outlets into City (codes 11–13), Suburban (21–23), Town (31–33), and Rural (41–43) based on proximity to urbanized areas—has been particularly useful for spatial comparisons, aligning with federal spatial policy frameworks [9].

### *D. Research Gap*

Prior work has established that urban-rural service gaps exist, but most studies focus on specific dimensions or use library-system-level data rather than individual outlet records. The simultaneous relationships among physical footprint, operating schedule, outlet classification, and population context—at the full national outlet scale using the most recent pre-pandemic data—remain uncharacterized. This study addresses that directly.

## III. DATASET AND METHODOLOGY

### *A. Data Source*

This study uses the FY 2018 Public Libraries Survey Outlet Data file (pls_fy18_outlet_pud18i), obtained from IMLS. The dataset was collected by the American Institutes for Research using the PLS Web Portal, covering the fiscal year ending June 30, 2018 [9]. It contains one row per physical point of service—17,478 observations across 37 variables—including state code (STABR), outlet type (C_OUT_TY), square feet (SQ_FEET), annual operating hours (HOURS), weeks open (WKS_OPEN), NCES locale code (LOCALE), and county population (CNTYPOP). This is distinct from the administrative entity (AE) system-level file.

### *B. Data Preprocessing and Feature Engineering*

Preprocessing ran in three stages. First, duplicate rows were checked and none found—expected, since each outlet has a unique FSCSKEY–FSCS_SEQ identifier. Second, records with negative placeholder values (used by IMLS to flag non-response or inapplicable numeric fields) were removed for analyses involving SQ_FEET, HOURS, and WKS_OPEN, leaving a clean analytical subset of roughly 17,050 outlets. Third, a derived variable was created: HOURS_PER_WEEK, calculated as annual hours divided by weeks open, giving a normalized weekly access measure that does not conflate reduced weekly hours with seasonal closures.

Categorical variables were recoded. Outlet type (C_OUT_TY) was mapped to four categories: Central (CE), Branch (BR), Bookmobile (BS), and Books-by-Mail Only (MO). NCES locale codes were aggregated from 12 subcategories to four primary types—City, Suburb, Town, and Rural—consistent with prior PLS research [7], [9].

### *C. Analytical Methods*

Analysis used a multi-method exploratory and inferential approach. Descriptive statistics (counts, means, medians, standard deviations) covered all key variables. Distribution analysis used histograms with kernel density overlays. Cross-tabulation and grouped aggregation compared outlet type and locale proportions. Box plots and violin plots characterized distributional differences across groups. Pearson's r correlation was computed across four numeric variables (SQ_FEET, HOURS_PER_WEEK, WKS_OPEN, CNTYPOP) and displayed as an annotated heatmap. Population-stratified analysis divided county population into five bins (<25K, 25K–100K, 100K–500K, 500K–1M, >1M) and computed mean annual hours per stratum. Regression visualization used a linear trend line with scatter on a logarithmic population scale. All analysis was done in Python with pandas, matplotlib, seaborn, and numpy.

## IV. RESULTS AND ANALYSIS

### *A. National Overview*

The FY 2018 outlet file covers 17,478 library outlets across all 50 U.S. states, D.C., and four outlying territories. Central libraries are the most common type (n = 9,092; 52.0%), followed by Branch (n = 7,757; 44.4%), Bookmobile (n = 628; 3.6%), and Books-by-Mail Only (n = 1; <0.1%). Together, Central and Branch libraries account for 96.4% of all outlets. Among outlets with valid

square footage data, median facility size is 6,550 sq. ft. and mean is 12,028 sq. ft., with a maximum of 970,000 sq. ft.—the distribution is heavily right-skewed, pulled by a small number of very large urban central libraries. Mean annual operating duration is 2,174 hours (~41.8 hours per week), and the average number of weeks open per year is 51.4. More than 82% of outlets operate all 52 weeks.

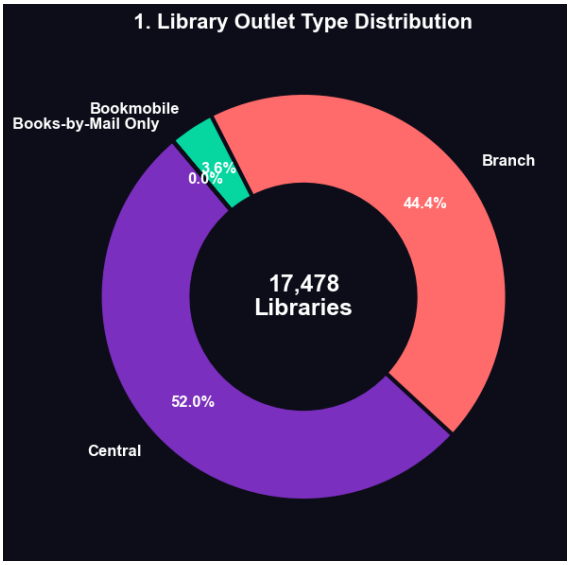


*Fig. 1. Donut chart showing proportional distribution of U.S. public library outlets by type (FY 2018, n = 17,478). Central and Branch libraries together account for 96.4% of the national outlet inventory.*

Fig. 1 shows the outlet type breakdown. The dominance of Central libraries (52%) reflects how U.S. library systems are typically organized—one central outlet anchors the system and houses the main collection and administrative staff, while branch libraries extend geographic reach into neighborhoods. Bookmobiles (3.6%) represent a service model designed for areas where building a fixed facility is not economically viable, especially in rural and remote communities [14].

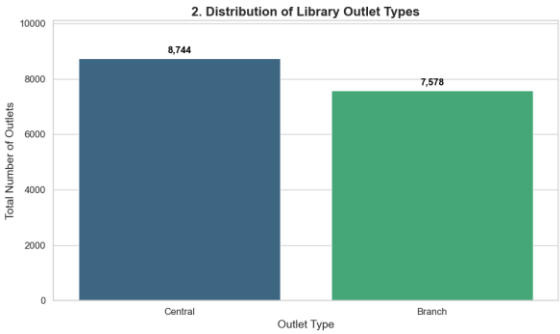


*Fig. 2. Bar chart showing absolute outlet counts by type (FY 2018).*

***B.*** *Geographic Distribution of Library Outlets*

California (n = 1,225), New York (n = 1,080), and Texas (n = 896) together account for roughly 18.9% of all outlets nationwide. This reflects state population size, library funding laws, and the density of municipal jurisdictions that independently run library systems. New York stands out: its decentralized municipal governance structure produces an unusually high outlet-to-population ratio compared to similarly sized states [12].

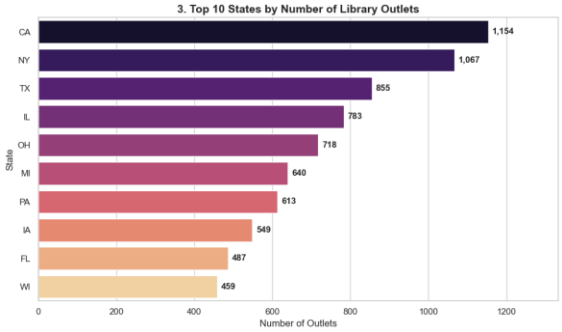


*Fig. 3. Horizontal bar chart of the top 10 U.S. states by outlet count (FY 2018). California, New York, and Texas lead nationally.*

Suburban areas host the largest share of library outlets nationwide, ahead of City, Rural, and Town settings (Fig. 4). This suburban predominance traces back to post-World War II library expansion into growing suburban municipalities, driven by population growth, stronger property tax bases, and state library development programs [3]. That the majority of the national library footprint is concentrated in suburban and city settings carries clear implications for spatial equity.

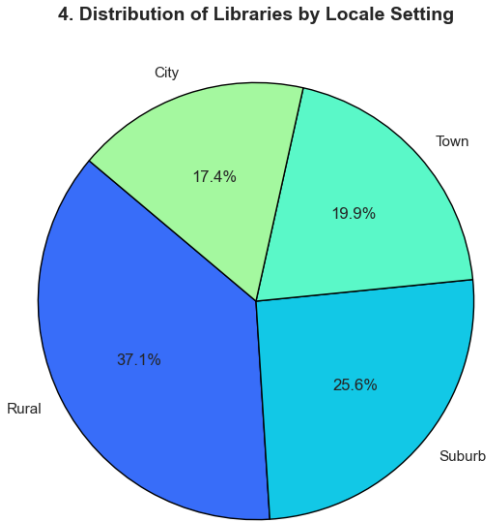


*Fig. 4. Pie chart of U.S. library outlets by NCES locale category (FY 2018). Suburban areas host the largest share nationally.*

Fig. 5 shows the infrastructure mix within each locale type. City settings have a notably higher proportion of Branch libraries—reflecting the multi-branch model used by large urban systems to serve dense populations. Rural settings show the highest relative share of Bookmobiles, an adaptive response to geographic distances that make fixed-facility coverage prohibitively expensive at low population density [10], [14].

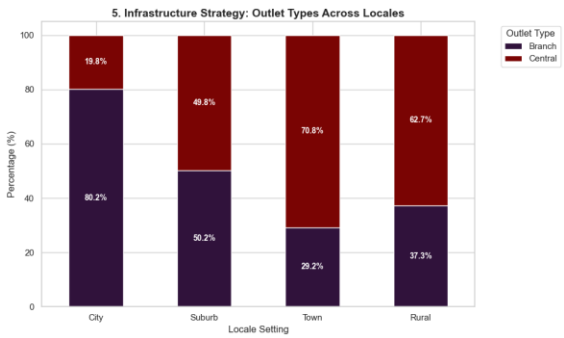


*Fig. 5. Stacked bar chart of outlet type composition within each locale category (FY 2018). Rural areas have the highest relative Bookmobile share.*

### *C. Physical Facility Size*

The facility size distribution (Fig. 6, filtered to ≤50,000 sq. ft.) is strongly right-skewed, with the median well below the mean. Most outlets fall between 2,000 and 8,000 sq. ft., consistent with IMLS data showing that the average outlet serving fewer than 1,000 people measures roughly 2,151 sq. ft. [9]. The kernel density overlay shows a bimodal pattern: a primary mode around 3,000–4,000 sq. ft. (small branches and rural centrals) and a secondary mode around 8,000–12,000 sq. ft. (mid-size central and large branch libraries). A smaller tail extends into the 30,000–50,000 sq. ft. range, where large urban central libraries sit.

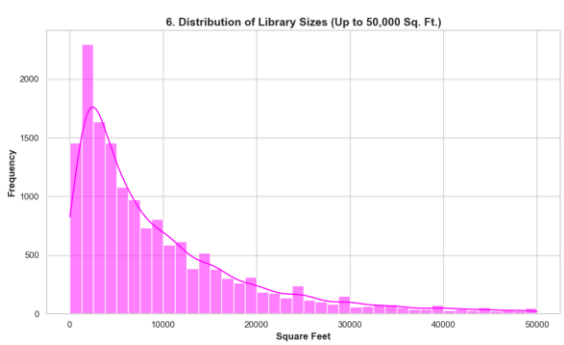


*Fig. 6. Histogram with KDE overlay of library facility sizes (≤50,000 sq. ft.). Primary mode falls at 3,000–4,000 sq. ft.*

The locale-stratified box plot (Fig. 7) shows real differences in facility size across locale types. City libraries have the highest median size and the widest interquartile range—mixing very large centrals with small neighborhood branches. Suburban libraries are similarly large but less variable. Town and Rural libraries are progressively smaller with tighter ranges, indicating that rural systems have converged around a more uniform and smaller footprint. This is consistent with Sin's earlier finding that urbanization level is a significant predictor of outlet size [8].

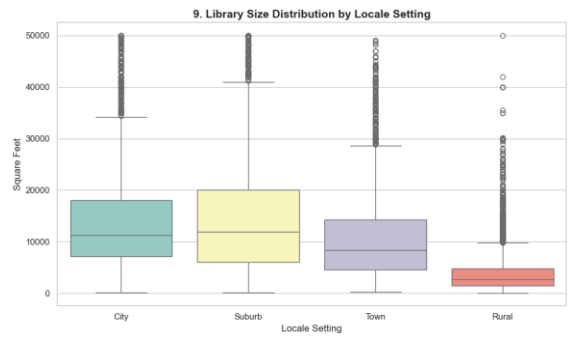


*Fig. 7. Box plot of facility size (sq. ft.) by NCES locale category (FY 2018, ≤50,000 sq. ft.). City libraries show the highest median and most variation.*

### ***D.*** *Operating Hours: Distributions and Outlet-Type Patterns*

Annual operating hours (Fig. 8) concentrate in the 1,500–3,000 hours range, with a mean of 2,174 hours and a maximum of 5,526 hours. A substantial share of outlets operate in the 2,000–2,500 hour band, consistent with a roughly 40-hour-per-week, 52-week schedule. Some outlets exceed 4,000 hours per year (~77 hours per week), pointing to extended-hours service models in high-demand urban settings.

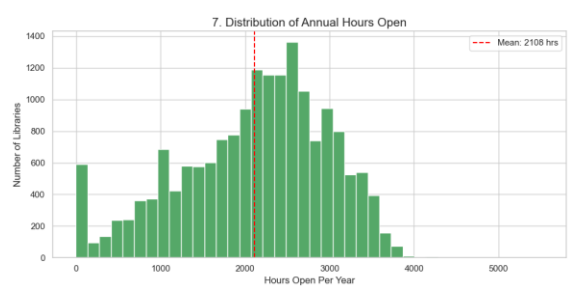


*Fig. 8. Histogram of annual operating hours for all U.S. public library outlets (FY 2018). Modal range is 1,500–3,000 hours per year.*

Fig. 9 compares average weekly hours by outlet type. Central libraries average about 44.2 hours per week; Branch libraries average about 40.1 hours. Bookmobiles average considerably fewer—they follow circuit routes rather than a continuous-presence model, which is a fundamentally different operating structure. The violin plot in Fig. 10 adds distributional detail: Central library hours cluster tightly in the 35–50 hour range, while Branch library hours are more spread out, reflecting the greater variety in branch size and neighborhood context.

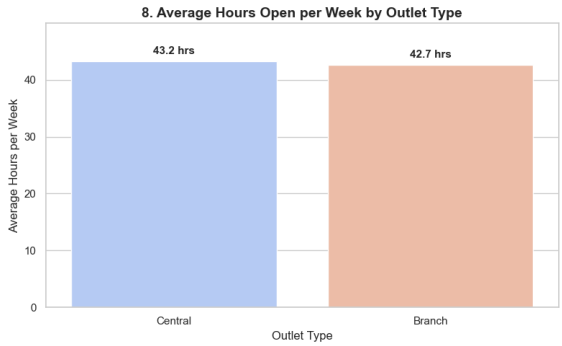


*Fig. 9. Bar chart of average weekly operating hours by outlet type (FY 2018). Central libraries average 44.2 hours per week.*

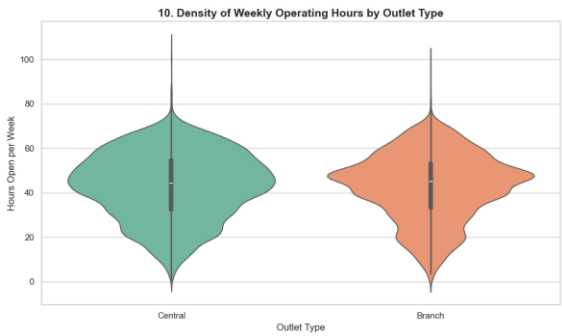


*Fig. 10. Violin plot of weekly operating hours by outlet type (FY 2018). Central libraries show a tighter, higher concentration of hours than Branch libraries.*

### *E. Correlation Analysis*

Fig. 11 shows Pearson correlation coefficients across four variables: facility square footage, weekly operating hours, weeks open per year, and county population. The strongest correlation is between SQ_FEET and HOURS_PER_WEEK ($r \approx 0.44$), indicating that larger facilities generally stay open longer. This makes sense: larger buildings serve larger populations with more demand for access time. The correlation between CNTYPOP and HOURS_PER_WEEK is also moderately positive ($r \approx 0.39$), suggesting that libraries in more populous counties run longer weekly hours—likely because both patron demand and local tax revenue support extended operations [7], [15]. WKS_OPEN shows weaker correlations across the board, which is expected given that more than 82% of outlets operate all 52 weeks, leaving little variance to explain.

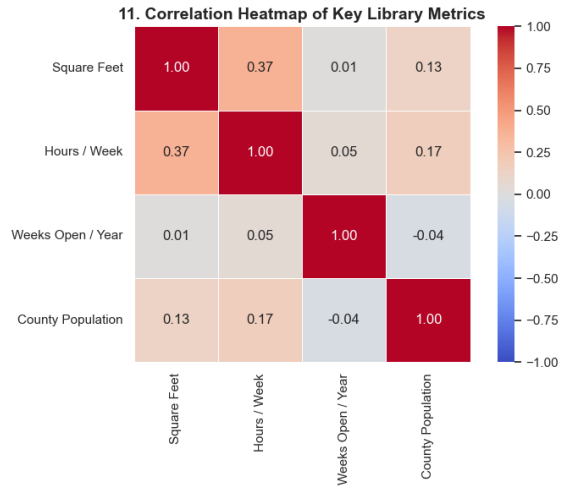


*Fig. 11. Pearson correlation heatmap among key library metrics: square footage, weekly hours, weeks open, and county population (FY 2018).*

### *F. County Population and Operational Capacity*

Fig. 12 plots mean annual operating hours by county population bracket. The pattern is monotonically positive: libraries in counties below 25,000 residents average about 1,780 hours per year (~34.2 hours per week), while libraries in counties above one million residents average about 2,690 hours per year (~51.7 hours per week). That gap of roughly 910 annual hours equals more than 22 full operational weeks. The curve steepens between the 25K–100K and 100K–500K bins, suggesting the shift from small-town to mid-size community contexts is where service intensity increases most sharply.

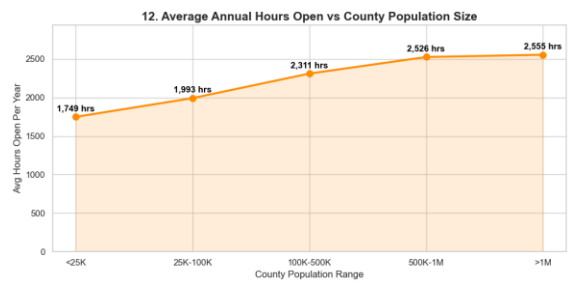


*Fig. 12. Line chart with area fill showing mean annual operating hours by county population bracket (FY 2018). Libraries in counties above 1 million average roughly 2,690 hours per year.*

The scatter-regression plot in Fig. 13 shows library size versus weekly operating hours, colored by outlet type. Central libraries cluster at higher values on both axes, while Bookmobiles sit at the lower end. Fig. 14 uses a logarithmic population scale to show the relationship between county population and facility square footage. The regression line has a clearly positive slope. Scatter around the trend is substantial—local funding decisions and planning choices introduce real variation—but the overall

direction is consistent and unambiguous. IMLS's own tabulation shows a similar gradient: the average outlet serving populations below 1,000 measures 2,151 sq. ft., compared to 19,750 sq. ft. for outlets serving 100,000 or more [9].

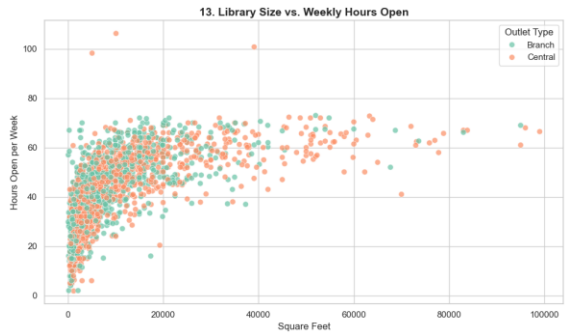


*Fig. 13. Scatter plot of library size (sq. ft.) versus weekly operating hours, colored by outlet type (sample of 2,000 outlets, FY 2018).*

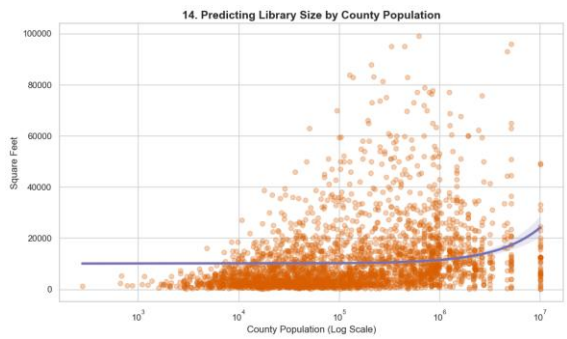


*Fig. 14. Regression scatter of county population (log scale) versus facility square footage (sample of 3,000 outlets, FY 2018). Positive association between population and physical capacity is evident.*

## V. DISCUSSION

### *A. The Urban-Rural Infrastructure Gap*

Taken together, the findings describe a consistent pattern: city libraries are larger, open longer, and better provisioned than rural ones. The gap cannot be explained by lower rural need—rural residents may be more dependent on their local library as a sole information access point, given lower rates of home broadband penetration [5], [10]. The gap traces to how libraries are funded. Property-tax-based local financing rewards communities with higher land values and bigger commercial tax bases, directing more dollars to already-advantaged suburban and urban areas [7], [11].

The finding that suburban areas host the most outlets nationally complicates a simple city-versus-rural story. Suburban library systems expanded through decades of post-war municipal growth and relatively stable tax revenues. But the size and hours data suggest suburban outlets, while numerous, tend to be smaller than city outlets—many mid-size facilities spread across a larger area rather than fewer large ones. This may be a rational spatial adaptation, but it also means any individual suburban branch has a smaller catchment and lower per-visit capacity than a large urban central library.

### *B. The Operating Hours Gap and Patron Access*

A 910-hour annual gap between the smallest and largest county population brackets has real consequences. A library open 34 hours per week reaches a different set of people than one open 52 hours per week—the shorter schedule may effectively exclude working adults who can only visit evenings and weekends. The finding that over 82% of outlets stay open all 52 weeks is reassuring in the year-round dimension, but the within-week hours shortfall remains a practical equity concern.

This is compounded by the correlation between operating hours and county population ($r \approx 0.39$): the communities with the fewest library hours also tend to be the smallest and most rural—exactly where alternative information resources are least available [5], [10]. Rural libraries may need to consider extended evening or weekend hours, collaborative service models with regional systems, or shared scheduling strategies to close that gap without proportional increases in staffing costs.

### *C. Bookmobiles as Adaptive Infrastructure*

The 3.6% of outlets that are Bookmobiles represent a practical answer to a funding and geography problem. Where per-capita tax revenue cannot sustain a fixed facility, a single staff team and vehicle can serve multiple communities on a rotating circuit. The higher relative Bookmobile share in rural locales (Fig. 5) reflects this logic directly. But recent work questions whether Bookmobiles adequately address digital inclusion: mobile units typically cannot provide sustained broadband access or technology training comparable to a fixed facility [10], [14]. The persistence of the Bookmobile model in FY 2018 data suggests the structural funding constraints that make it necessary have not eased much.

### *D. Limitations*

Several limitations should be noted. The analysis is cross-sectional, covering one fiscal year; it cannot track trends over time. The outlet-level data contain no patron visit counts, circulation figures, or program attendance, which would enable demand-adjusted capacity measures. The county population variable (CNTYPOP) is an imperfect proxy for a library's actual service area population, particularly where library boundaries do not align with county boundaries [9]. And square footage carries some ambiguity in multi-story buildings where only part of the space is publicly accessible. Future work should integrate

PLS AE-level financial data, multi-year panels, and census-tract-level socioeconomic variables.

## VI. CONCLUSION

This paper analyzed the FY 2018 IMLS Public Libraries Survey outlet-level data across 17,478 outlets in all U.S. states and territories. Four main findings stand out. First, Central and Branch libraries account for 96.4% of all outlets, with suburban areas hosting the most outlets nationally. Second, city libraries are consistently larger and open for more weekly hours than rural ones—the annual hours gap reaches about 910 hours between the smallest and largest county population brackets. Third, county population has a moderate positive correlation with both facility size and operating hours, pointing to a demand-capacity alignment that may nonetheless leave high-need rural communities short. Fourth, facility size and weekly hours are moderately correlated, suggesting that physical investment in buildings tends to come with matching operational investment.

These results give researchers a quantitative baseline—at outlet resolution, using the last full pre-pandemic data year—against which post-COVID library recovery can be measured. For policymakers, they argue for looking beyond internet access metrics when assessing library equity. Building size and operating hours are concrete dimensions of service that shape whether a library can actually function as a community information hub. Narrowing the urban-rural gap in these dimensions deserves attention alongside connectivity investments.

Future research should extend this analysis across multiple PLS waves to see whether the gap has narrowed or widened, incorporate AE-level financial data to model funding mechanisms as mediators, and explore how national patterns hold up—or break down—within specific regional or state contexts. Linking the data to census-tract socioeconomic and health outcomes would allow researchers to move from describing infrastructure gaps to estimating the community-level consequences of closing them.

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